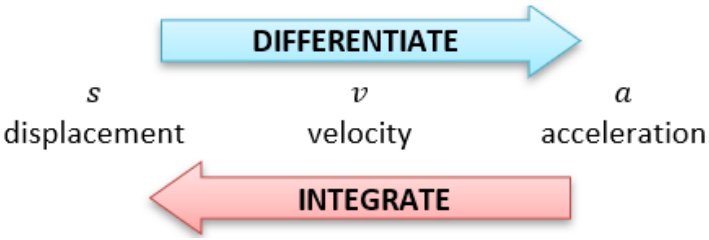
1. General Motion in a Straight Line



- Particle at instantaneous rest, $v = 0$
- Maximum displacement from origin, $v = 0$
- Maximum velocity, $a = 0$

{W10-P42} Question 7:

Particle P travels in straight line. It passes point O with velocity $5ms^{-1}$ at time $t = 0s$.

P's velocity after leaving O given by:

$$v = 0.002t^3 - 0.12t^2 + 1.8t + 5$$

v of P is increasing when: $0 < t < T_1$ and $t > T_2$

v of P is decreasing when: $T_1 < t < T_2$

i) Find the values of T_1 and T_2 and distance OP when $t = T_2$

ii) Find v of P when $t = T_2$ and sketch velocity-time graph for the motion of P

Solution

Part (i)

Find stationary points of v; maximum is where $t = T_1$ and minimum is where $t = T_2$

$$\frac{\mathrm{dv}}{\mathrm{dt}} = 0.006t^2 - 0.24t + 1.8$$

Stationary points occur where $\frac{\mathrm{dv}}{\mathrm{dt}} = 0$

$$\therefore 0.006t^2 - 0.24t + 1.8 = 0$$

Solve for t in simple quadratic fashion:

$$t = 30 \quad and \quad 10$$

Naturally T_1 comes before T_2

$$\therefore T_1 = 10s \qquad and \qquad T_2 = 30$$

Finding distance OP by integrating

$$\therefore s = \int_0^{30} v\mathrm{dt}$$

$$s = \int_0^{30} \left(0.002t^3 - 0.12t^2 + 1.8t + 5\right) \mathrm{dt}$$

$$s = \left[0.0005t^4 - 0.04t^3 + 0.9t^2 + 5t\right]_0^{30}$$

$$s = 285\ m$$

Part (ii)

Do basic substitution to find v

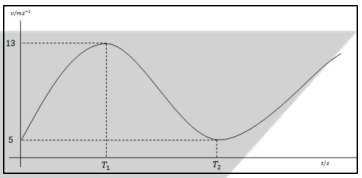
$$v = 0.002t^3 - 0.12t^2 + 1.8t + 5$$

$$t = 30 \qquad v = 5$$

To draw graph, find v of P at T_1 using substitution and plot roughly

$$\text{v at } T_1 = 13$$

Graph:



{S13-P42} Question 6:

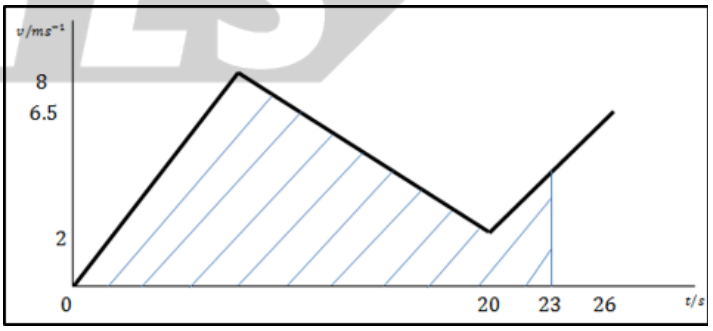
Particle P moves in a straight line. Starts at rest at point O and moves towards a point A on the line. During first 8 seconds, P's speed increases to $8ms^{-1}$ with constant acceleration. During next 12 seconds P's speed decreases to $2ms^{-1}$ with constant deceleration. P then moves with constant acceleration for 6 seconds reaching point A with speed $6.5ms^{-1}$

1. Sketch velocity-time graph for P's motion
2. The displacement of P from O, at time t seconds after P leaves O, is s metres. Shade region of the velocity-time graph representing s for a value of t where $20 \le t \le 26$
3. Show that for $20 \le t \le 26$,

$$s = 0.375t^2 - 13t + 202$$

Solution:

Part (i) and (ii)



Part (ii)

First find s when $t = 20$, this will produce a constant since $20 \le t \le 26$

$$s_1 = \frac{1}{2}(8)(8) + \frac{1}{2}(8+2)(12) = 92m$$

Finding s when $20 \le t \le 26$:

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$$s = ut + \frac{1}{2}at^2$$

Since the distance before 20 seconds has already been taken into consideration:

$$\mathbf{t = t - 20}$$

$$a = \frac{6.5 - 2}{6}$$

$$a = 0.75$$

$$\therefore s_2 = 2\left(t - 20\right) + \frac{1}{2}\left(0.75\right)\left(t - 20\right)^2$$

$$s_2 = 2t - 40 + 150 + 0.375t^2 - 15t$$

$$s_2 = 0.375t^2 - 13t + 110$$

Finally, add both to give you s

$$s = s_1 + s_2$$

$$s = 0.375t^2 - 13t + 110 + 92$$

$$s = 0.375t^2 - 13t + 202$$

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